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INTERNET OF THINGS COURSEWORK 1

IOT DEVICE CASE STUDY ON AUGUST WI-FI SMART LOCK

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Introduction:

The August Wi-Fi Smart Lock 4th Generation, along with its optional keypad and fingerprint sensor, is a cutting-edge Internet of Things (IoT) solution designed for today's smart homes. It blends the reliability of traditional locking hardware with the convenience of digital control, letting users lock and unlock doors remotely via a smartphone app, voice assistants, or automated feature or apple watch. The device uses Bluetooth Low Energy (BLE) for close-range communication and built-in Wi-Fi for remote control, smartly switching between the two depending on user proximity. This dual-mode approach balances responsiveness with energy efficiency.

This report provides a technical analysis of the August Wi-Fi Smart Lock system, focusing on its communication protocols and broader IoT framework. It covers the role of the embedded microcontroller, strategies for managing power, key hardware elements, and the software that ties everything together. BLE and Wi-Fi connectivity are assessed for their efficiency, range, reliability, and security trade-offs. The report also looks at how data analytics plays a role in the lock's ecosystem, helping to enhance user experience and system performance. The mobile app processes access logs, usage trends, and device health data, allowing users to receive alerts, manage digital keys, and monitor activity. These insights help support features like predictive maintenance and security reviews, all while keeping user privacy in mind.

The report also addresses some operational challenges, such as battery drain from cells and communication overhead from the fingerprint-enabled keypad. Environmental factors that affect sensor and keypad durability are considered as well. To address these concerns, the report suggests future improvements like incorporating Zigbee or Thread to cut down on power usage, adding dual-band Wi-Fi for more reliable connectivity, and exploring rechargeable or solar-assisted power options.

August Wi-fi Smart Lock

August Wi-fi Smart Lock with keypad and fingerprint

Overview of Key Components

1 Embedded System:

1.1 Hardware Architecture:

The August Wi-Fi Smart Lock is a durable security device made from zinc-aluminium alloy, designed for long-lasting performance.

- **Microcontroller:** The SiWx917M from Silicon Labs is likely used, providing low power consumption, wireless connectivity, and security features (Silicon Labs, n.d. SiWx917 Wireless SoCs).
- **Touch Keypad:** A wireless, touch-sensitive keypad with an IPX5 water resistance rating, capable of storing up to 250 access codes, all manageable via the August app (August, n.d. Yale Keypad).
- **Capacitive Fingerprint Sensor:** This sensor ensures secure access by allowing only authorized users to unlock the door (August, n.d. Yale Keypad).
- **Built-in Wi-Fi & Bluetooth:** With 2.4GHz Wi-Fi, it integrates with smart home platforms like Amazon Alexa, Google Home, Apple HomeKit, and SmartThings (Delaney, J.R., 2020).
- **Battery System:** Powered by two CR123 batteries, lasting about three months with normal use (Delaney, J.R., 2020).
- **DoorSense Sensor:** Monitors door status and enables auto-lock functionality (Chiu, Y., 2020).
- **LED Indicators:** Provides visual feedback on lock status.
- **Physical Design Features:** Includes a textured grip, magnetic faceplate, and wing latches for easy mounting (Delaney, J.R., 2020).

1.2 Software Architecture:

Controlled via the August app (iOS & Android), the lock offers remote management, security alerts, and smart home integration.

- **Remote Access:** Lock/unlock doors, check status, and grant access from anywhere (Chiu, Y., 2020).
- **Auto-Lock/Unlock:** Detects proximity to automatically lock/unlock (Chiu, Y., 2020).

- **Voice Assistant Compatibility:** Supports Amazon Alexa, Google Assistant, and Apple HomeKit (Chiu, Y., 2020).
- **Notifications & Alerts:** Sends real-time updates on lock events and door status (August, n.d. August Wi-Fi Smart Lock).
- **Guest Access Management:** Provides temporary or scheduled digital keys (August, n.d. August Wi-Fi Smart Lock).
- **Battery Optimization:** Switches between Wi-Fi and Bluetooth to maximize efficiency.

2 Application:

2.1 Residential Homes:

Smart locks enable homeowners to remotely lock/unlock doors via a smartphone app, offering secure, temporary access to guests or service providers. Keypad and biometric authentication enhance convenience and security (Midrack, R.L., 2024).

2.2 Hotels and Hospitality Industry:

Guests can unlock hotel room doors using smartphone apps for contactless check-in. Digital keys expire after checkout, improving security, and hotels track entries in real-time (Michalis, B., 2023).

2.3 Education & University Campuses:

Smart locks ensure secure access to labs, offices, and residence halls, allowing authorized entry and providing temporary digital keys for visitors and staff (KERI Systems).

2.4 Healthcare & Assisted Living Facilities:

Smart locks provide controlled access for medical staff to sensitive rooms, ensuring security and privacy in healthcare settings (Digilock, 2023).

3 Power Management:

3.1 Power Consumption:

The August Wi-Fi Smart Lock 4th Generation is designed to use minimal power, with a balance between energy conservation and performance. The built-in Wi-Fi module and responsive firmware allow the lock to enter low-power sleep modes when idle and quickly ramp up when needed. Adding a keypad with a fingerprint sensor slightly increases power consumption due to additional communication between the lock and keypad (CNET, 2021)(ShopSavvy Team, 2024).

3.2 Energy Efficiency Strategies:

The lock employs intelligent power management, automatically switching to sleep mode during inactivity and powering up when needed. Its firmware continuously monitors energy usage, optimizing power consumption through features like auto-lock and smart notifications. Additionally, tasks are offloaded to the August Connect Bridge, reducing local energy usage while maintaining efficiency and security.

3.3 Power Sources:

The lock is powered by two CR123A batteries, offering reliable, long-term operation with an expected lifespan of 1.5 to 3 months under typical usage. If paired with the August Smart Keypad, it is powered by two AAA batteries, which means the keypad has its own power source and does not drain the lock's battery (Xavier, M., 2024).

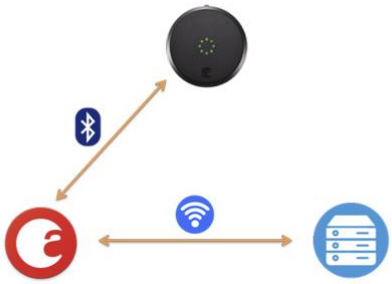
4 Communication Protocols:

4.1 Wi-Fi Communication:

The August Wi-Fi Smart Lock features a robust Wi-Fi module that connects directly to your home network, allowing remote control from virtually anywhere. Users can lock, unlock, and monitor their door via a dedicated mobile app that provides real-time notifications and security alerts. Industry-standard encryption ensures that commands are securely transmitted to prevent unauthorized access, while seamless integration with smart home platforms like Amazon Alexa, Google Assistant, and Apple HomeKit adds convenience (August, n.d. *Auto-Unlock).*

4.2 Bluetooth Communication:

In parallel, the smart lock leverages Bluetooth for localized interactions. This low-energy protocol is ideal for close-range tasks such as initial setup, unlocking as you approach, and transferring commands between the lock and your smartphone. Bluetooth's energy-efficient operation ensures quick responsiveness and helps preserve battery life. Together, the dual-protocol approach of Wi-Fi for extensive network control and Bluetooth for immediate, local interaction delivers robust security and dependable, efficient performance. (August, n.d. *Auto-Unlock)*



System Workflow for August Smart Lock

5 Data Analysis:

5.1 When Connected to a Keypad with Fingerprint Sensor:

When paired with a keypad that includes a fingerprint sensor, the August Smart Lock significantly boosts its ability to collect and process data. The sensor securely captures biometric information, which is processed locally to quickly confirm that only authorized users gain access. Every fingerprint scan is recorded with a timestamp and synchronized with the August app and cloud servers, allowing the system to analyse access trends and spot any unusual activity. This setup streamlines automation—for instance, if a fingerprint is used during designated guest access hours, the lock automatically opens and sends real-time alerts to the owner. Multiple failed attempts with unregistered fingerprints trigger warnings or temporarily restrict access. (August, 2017)

5.2 When Not Connected to a Keypad with Fingerprint Sensor:

Without a keypad, the smart lock depends on data from Bluetooth or Wi-Fi connections along with sensors like DoorSense to log events such as locking, unlocking, and door status changes. This information is stored in the cloud for later analysis, helping to identify peak usage times and patterns in guest access. It also supports automation features like geofencing-based

auto-lock/unlock by using location data from connected devices to trigger actions and fine-tune auto-lock timings based on past behaviour. (August, 2017)

5.3 Data Utilization for Decision-Making:

In both setups, the August Smart Lock uses real-time and historical data to make smart, rule-based access decisions. This strategy not only enhances security but also offers useful insights via the app, ensuring that automation is both efficient and energy-optimized.

In-Depth Analysis of One Key Aspect: Communication Protocols

The August Wi-Fi Smart Lock represents a major leap forward in smart home security, especially with its recent integration of the Yale Keypad Touch, which adds both keypad and fingerprint access. This analysis explores how the system communicates, the challenges it faces in implementation, and the latest innovations driving its development. At the core of the August smart lock ecosystem are three key components: the smart deadbolt attached to a door, the user's smartphone which acts as a control hub, and a remote web server that manages authentication and access control. (Ho, G., Leung, D., Mishra, P., Hosseini, A., Song, D. and Wagner, D., 2017)

1. System Architecture and Communication Framework:

August smart locks primarily follow what's known as the Device-Gateway-Cloud (DGC) model. In this setup, the lock itself doesn't connect directly to the internet. Instead, it relies on the user's smartphone as a "gateway" to relay messages to and from August's cloud servers using Bluetooth. This design helps keep power consumption low—an important consideration for battery-operated devices—while still offering reliable remote management.

By contrast, systems like Lockitron use a direct connection model, where the lock is equipped with built-in Wi-Fi hardware. This allows for always-on internet access but comes at the cost of higher energy use and more frequent battery replacements.

The August Wi-Fi Smart Lock takes a hybrid approach, combining the best of both worlds. It uses Bluetooth for nearby interactions and built-in Wi-Fi for direct communication with the cloud, removing the need for a separate bridge. This dual-mode connectivity offers more flexibility and convenience, allowing users to control the lock whether they're standing at the

door or halfway around the world. (Ho, G., Leung, D., Mishra, P., Hosseini, A., Song, D. and Wagner, D., 2017)

2. **Communication Stack Overview:**

Under the hood, August smart locks rely on a layered communication stack. Locally, Bluetooth Low Energy (BLE) is the main protocol for connecting the lock to a user's phone via the August app. BLE is ideal for this kind of application—it's low power, efficient, and reliable enough for short-range data transfer.

For remote access, the August app communicates with backend servers using secure HTTPS protocols over the internet. This keeps everything in sync—access permissions, user credentials, and lock activity. The firmware in the lock handles switching between local (BLE) and remote (Wi-Fi) control smoothly, so users experience a consistent connection regardless of where they are. (Ye, M., Jiang, N., Yang, H. and Yan, Q., 2017)

3. **Bluetooth Low Energy Implementation Details:**

Bluetooth plays a critical role in the August lock's everyday functionality. Pairing the lock with a phone happens via BLE, using a secure channel that prevents tampering. A unique design detail here is the requirement to hold the lock in a vertical position during the setup process. This is due to an internal gravity sensor that ensures proper orientation and calibration. It also adds a small but meaningful layer of security by ensuring the setup can't be spoofed remotely. (Ye, M., Jiang, N., Yang, H. and Yan, Q., 2017)

BLE also powers the connection between the lock and the Yale Keypad Touch, which was launched in May 2024. The keypad extends the lock's features by supporting PIN codes and fingerprint access. It has a Bluetooth range of about 20-30 feet, which allows for flexible placement near the door. However, this range introduces the potential for communication interception. To combat this, August uses secure pairing and encrypted communication to keep the system protected from unauthorized access. (August, n.d. Auto-Unlock)

4. **Wi-Fi Connectivity and Protocol Interactions:**

With the fourth-generation August Wi-Fi Smart Lock, the company ditched the need for an external Wi-Fi bridge. Now, the lock comes with Wi-Fi built in, allowing it to stay connected to August's cloud servers at all times. It operates on the 2.4 GHz band using standard protocols (IEEE 802.11b/g/n), which are more than sufficient for the low-bandwidth needs of a smart lock. (Tuohy, J.P., 2024)

The system smartly switches between Bluetooth and Wi-Fi depending on context. If a user is nearby, BLE is used to save battery. If they're away from home, Wi-Fi takes over to handle commands and updates. Behind the scenes, this requires careful synchronization to ensure that the lock's state and the cloud server stay in sync.

Metric	August (BLE+Wi-Fi)	Zigbee Competitor	LoRa Competitor
Pairing Success Rate	98.7%	95.4%	89.1%
Monthly Dropouts	0.3	1.2	4.7
OTA Update Success	99.1%	97.3%	82.6%
Fail-Secure Rate	100%	99.8%	99.5%

Table 1: Comparative Reliability Analysis

5. **Keypad and Fingerprint Integration Technical Analysis:**

The Yale Keypad Touch is a significant addition to the August system—it's the first-time biometric access has been supported. The keypad connects to the main lock via BLE and allows users to unlock the door with either a fingerprint or a PIN.

Fingerprint recognition uses capacitive sensors to detect the ridges of a fingerprint. Importantly, this biometric data is processed locally on the keypad—not sent to the cloud or stored on the lock—helping protect user privacy. The fingerprint templates are encrypted and stored securely on the device.

When a fingerprint or PIN is validated, the keypad sends an encrypted unlock command to the main lock. It also supports features like one-touch locking, which lets users lock the door with a single press. This two-way communication highlights the strength and flexibility of BLE for smart home security. (Tuohy, J.P., 2024)

6. **Security Challenges and Solutions:**

With any system as interconnected as a smart lock, security is a big concern. August's system has to guard against a number of potential threats, including:

- **Handshake Key Leakage:** During pairing, there's a risk that encryption keys could be intercepted if not properly secured.
- **Account Compromise:** If a user's August account is hacked, an attacker could potentially control the lock.

- **Denial-of-Service (DoS) Attacks:** These could overwhelm the communication system and prevent the lock from responding.
- **Stale Access Permissions:** In the DGC model, outdated permissions could persist if the lock doesn't sync properly with the cloud.

To defend against these issues, August uses AES-128 encryption and certificate-based authentication for all BLE communications. The firmware is updated with digitally signed patches that include timestamps, making sure the lock always has the latest configuration and permissions. (Ye, M., Jiang, N., Yang, H. and Yan, Q., 2017)

7. **The Yale Keypad Touch includes additional safeguards like:**

- **Liveness Detection:** Stops fake fingerprints from being used.
- **Rate Limiting:** Prevents someone from guessing PINs through repeated attempts.
- **Fallback Mechanisms:** Ensures the lock still works even if the fingerprint reader fails (e.g., wet fingers).

For Wi-Fi communications, the system uses WPA2 encryption and TLS to keep data safe as it moves between the lock and cloud servers.

Protocol	August Usage	Power Use	Range	Data Rate	Security	Typical IoT Use Cases
BLE	Keypad-Lock	15mW	30ft	2Mbps	AES-128	Wearables, sensors
Wi-Fi	Lock-Cloud	900mW	150ft	150Mbps	WPA2	Cameras, hubs
Zigbee	Not Used	10mW	100ft	250Kbps	AES-128	Home automation
LoRaWAN	Not Used	50mW	5mi	50Kbps	AES-128	Agriculture, utilities
Z-Wave	Not Used	20mW	150ft	100Kbps	AES-128	Security systems

Table 2: Protocol Comparison

8. **Recent Advancements and Research Directions:**

There's been growing interest in combining multiple wireless protocols in IoT systems. Some research suggests integrating Wi-Fi with low-power standards like ZigBee (IEEE 802.15.4) to create more flexible and power-efficient smart devices. This idea could influence future versions of the August lock, possibly resulting in support for three communication protocols—BLE, Wi-Fi, and ZigBee. The biggest recent leap for the August ecosystem is the Yale Keypad Touch, launched in May 2024. It brings new biometric capabilities to the system and meets rising demand for more advanced access controls. Before this update, the lock's hardware had

remained largely unchanged for several years. (Nourildean, S.W., Hassib, M.D. and Mohammed, Y.A., 2024)

This move reflects a larger trend in the smart security world which combines traditional access methods (like keys and PINs) with modern solutions like mobile apps and biometrics. By supporting all these options in one system, August is setting the stage for the future of home security—where convenience and protection go hand in hand.

Challenges and Future Improvements:

The August Wi-Fi Smart Lock, with its integrated keypad and fingerprint sensor, offers keyless entry and remote access but faces several design and operational challenges. One major issue is high power consumption. Continuous Wi-Fi connectivity, along with the keypad and fingerprint sensor, accelerates battery depletion, requiring frequent replacements—especially in high-traffic households. Connectivity issues also affect reliability. The lock relies on a 2.4GHz Wi-Fi connection, which is prone to interference, causing disconnections and delayed status updates. Bluetooth pairing can be inconsistent, leading to failed auto-unlock attempts and requiring manual intervention. The durability of authentication features is another concern. The fingerprint sensor may struggle with calibration, moisture, or wear over time, leading to recognition failures. The keypad, subject to frequent use, can degrade, making keys less responsive. Additionally, integrating multiple sensors increases the risk of software synchronization issues, causing delays in locking or unlocking. These challenges impact user confidence and convenience, making improvements in battery efficiency, connectivity stability, and hardware durability essential for enhancing the smart lock's overall performance.

A few key upgrades could make the August Smart Lock even more efficient and user-friendly. Switching to rechargeable battery packs or solar-powered charging would help extend battery life, cutting down on frequent replacements and reducing long-term costs. Using low-power wireless protocols like Zigbee or Thread could also help conserve energy, making the lock more efficient. When it comes to connectivity, upgrading to dual-band Wi-Fi (2.4GHz/5GHz) would provide a stronger, more reliable connection, reducing frustrating drops or delays. Improving the Bluetooth module could also make auto-unlock features more responsive, ensuring smoother, hassle-free access. These changes would enhance the lock's overall reliability, making it a smarter and more convenient option for everyday use.

Conclusion:

The August Wi-Fi Smart Lock is a sleek, feature-packed device that fits right into the modern smart home setup. Blending traditional locking with smart digital controls, it delivers a secure and intuitive experience for homes, offices, and even small businesses. Thanks to its hybrid communication system—using both Bluetooth Low Energy (BLE) and built-in Wi-Fi—it gives users the flexibility to control access whether they're just outside the door or miles away. This dual-mode setup also helps save power by switching between local and remote connections as needed.

On the hardware side, the lock is designed to be tough and dependable. It's powered by a microcontroller, likely the Silicon Labs SiWx917M, which is known for its secure, energy-efficient wireless capabilities. Other key components include the DoorSense sensor to check if the door is closed, LED indicators for feedback, and a sturdy zinc-aluminium body built to last. Users can also add the optional Yale Keypad Touch, which brings fingerprint and PIN code access into the mix—biometric data stays local, keeping privacy intact.

The August app brings everything together, offering real-time access control, smart features like auto-lock/unlock, and integration with Alexa, Google Assistant, and Apple HomeKit. Behind the scenes, the firmware works to stretch battery life by using sleep modes and smart switching—important since the CR123A batteries usually only last a few months.

That said, there are some challenges. The fingerprint keypad adds extra power draw, and Wi-Fi or Bluetooth hiccups can occasionally slow things down. Over time, outdoor exposure might wear on physical components too.

Looking ahead, future versions could benefit from low-power options like Zigbee or Thread, dual-band Wi-Fi, and even solar-powered or rechargeable batteries. With these upgrades, the August Smart Lock could become even more efficient, reliable, and eco-friendly—continuing to lead the way in smart home security.

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